Cutaneous and Oral Lesions

Determining the ages of onset of skin lesions and their subsequent changes may assist in discriminating TSC skin lesions from similar-appearing non-TSC skin lesions. Many TSC skin lesions are not present at birth but appear later at ages typical for the lesion. For example, infants show multiple hypomelanotic macules but not angiofibromas or ungual fibromas. Adults may have multiple angiofibromas and ungual fibromas, but hypomelanotic macules may have faded or disappeared.

Hypomelanotic Macules

Hypomelanotic macules are observed in over 90 percent of children with TSC. They are often present at birth or appear within the first few years of life and may fade or disappear in adulthood. The ultraviolet light of a Wood's lamp is used to improve detection, especially in lightly pigmented individuals.

Hypomelanotic macules typically measure 0.5 to 3.0 cm in diameter. They are off-white and not completely depigmented as in vitiligo. Some are oval at one end and taper to a point at the other. Such lesions are called ash-leaf spots because of their resemblance to the leaf of the European mountain ash. They number from 1 to over 20 and can be located anywhere on the body but tend to occur on the trunk and buttocks. When located on the scalp, they cause poliosis.

Three or more hypomelanotic macules constitute a major feature for the diagnosis of tuberous sclerosis. One or two, and in rare individuals up to three, hypomelanotic macules occur in 4.7 percent of the general population. A less common type of hypopigmentation is the "confetti" skin lesion, which is considered a minor feature for diagnosis. It typically occurs on the legs below the knees or on the forearms and consists of multiple hypopigmented macules 2 to 3 mm in diameter.

Facial Angiofibromas

Angiofibromas appear at 2 to 5 years of age and eventually affect 75 percent to 90 percent of patients. These 1 to 3 mm pink to red papules have a smooth surface. They may be hyperpigmented, especially in individuals with darker skin tones. They occur on the central face and are often concentrated in the nasolabial folds, extending symmetrically onto the cheeks, nose, nasal opening, and chin, with relative sparing of the upper lip and lateral face. Sometimes lesions occur on the forehead, scalp, or eyelids. They may number from 1 to over 100. Lesions may coalesce to form large nodules, especially in the nasolabial folds.

The development of papules may be preceded by mild erythema that is intensified by emotion or heat. During puberty, angiofibromas may grow in size and number. During adulthood, they tend to be stable in size, but redness may gradually diminish.

Angiofibromas must be multiple to be counted as a major feature. A solitary angiofibroma is clinically and histologically indistinguishable from the fibrous papule that occurs sporadically as a single lesion in the general population. Multiple angiofibromas are also observed in multiple endocrine neoplasia type 1 and as an unusual finding in Birt-Hogg-Dubé syndrome.

Fibrous Facial Plaque

The fibrous facial plaque (forehead plaque) may be congenital or develop gradually over years. It is present in 20 percent to 40 percent of patients. It is an irregular, soft to firm, connective tissue nevus with the color of the normal surrounding skin, red, or hyperpigmented in darkly pigmented individuals.